

TOSE-Print: Mathematical Structure Tensor & Graph Topology Formulation

Dual-Branch Formulation: Continuous SE(2) Structure Tensor and Skeletal Homology Pipeline

Branch A: Continuous Structure Tensor on SE(2)

1. Spatial Gradient Estimation via Sobel Kernels:

$I_x = \partial I_{\text{prep}} / \partial x = I_{\text{prep}} * K_x, \quad I_y = \partial I_{\text{prep}} / \partial y = I_{\text{prep}} * K_y$
Spatial convolution with 3×3 separable Sobel operators

2. Scale-Integrated Structure Tensor Matrix J:

$J(x, y) = G_\sigma * [I_x^2, I_x I_y; I_x I_y, I_y^2] = [J_{xx}, J_{xy}; J_{xy}, J_{yy}]$
Gaussian integration scale $\sigma = 2.0$ over local contextual neighborhood
Local Principal Ridge Angle: $\theta(x, y) = \frac{1}{2} \text{atan2}(2J_{xy}, J_{xx} - J_{yy})$

3. Continuous Doubled-Angle Vector Field on SE(2):

$u_c(x, y) = \kappa \cos(2\theta) = \frac{J_{xx} - J_{yy}}{J_{xx} + J_{yy} + \epsilon}, \quad u_s(x, y) = \kappa \sin(2\theta) = \frac{2J_{xy}}{J_{xx} + J_{yy} + \epsilon}$
• **Continuous Norm**: $\|(u_c, u_s)\|_2 = \kappa(x, y) \in [0, 1]$ (Vanishes in untextured background)
• **Harmonic SO(2) Equivariance**: Rotation by $\phi \rightarrow 2\phi$ harmonic vector rotation

4. Sub-Vector Dimensions & Allocations:

Spatial Tensor Grid

128-d

8x8 regular grid (uc, us)

Weight: 0.55

Coherence Histogram

32-d

32-bin orientation polar

Weight: 0.20

Coherence Patch

64-d

8x8 macro-coherence map

Weight: 0.10

Branch B: Skeletal Graph Topology & Homology

1. Medial Axis Morphological Skeletonization:

Iterative morphological thinning to 1-pixel central ridge lines:
 $S(I) = \text{Skeletonize}(\mathcal{T}_{\text{Otsu}}(I_{\text{prep}})), \quad S(x, y) \in \{0, 1\}$
Preserves Euler characteristic $\chi = V - E + F$ and global homotopy tree

2. Rutovitz Crossing Number & Node Degree Analysis:

$CN(p) = \frac{1}{2} \sum_{i=1}^8 |p_i - p_{i+1}|, \quad p_9 \equiv p_1$ (8-neighborhood cycle)
• **Endpoints** ($CN = 1$): Normalized density $\rho_E = |E|/|V|$
• **Bifurcations** ($CN \geq 3$): Normalized density $\rho_B = |B|/|V|$

3. Homological Invariants & 1D Loop Spectra (H_0, H_1):

• **0 – th Betti Number** (β_0): Count of connected ridge graph components
• **1D Cycle Loop Homology** (H_1): Identifies closed ridge contours and islands
Normalized 32-bin cycle perimeter histogram: $\mathbf{v}_{\text{loop}} \in \mathbb{R}^{32}$, range $\in [0, 200]$ px
Topological summary invariant: $\mathbf{v}_{\text{topo}} = [\beta_0, \rho_E, \rho_B, N_{\text{loop}}, \bar{L}]^T \in \mathbb{R}^5$

4. Sub-Vector Dimensions & Allocations:

Loop Perimeter Hist

32-d

32-bin cycle lengths

Weight: 0.10

Topological Summary

5-d

[β_0 , ρ_E , ρ_B , N, L]

Weight: 0.05

Balanced Sub-Block Unit L_2 Normalization: $\mathbf{v}_{\text{TOSE}} = [\sqrt{0.55} \hat{\mathbf{v}}_{\text{spatial}} (128\text{d}) \parallel \sqrt{0.20} \hat{\mathbf{v}}_{\text{ori}} (32\text{d}) \parallel \sqrt{0.10} \hat{\mathbf{v}}_{\text{loop}} (32\text{d}) \parallel \sqrt{0.05} \hat{\mathbf{v}}_{\text{topo}} (5\text{d}) \parallel \sqrt{0.10} \hat{\mathbf{v}}_{\text{patch}} (64\text{d})]^T \in \mathbb{R}^{261}, \quad \|\mathbf{v}_{\text{TOSE}}\|_2 = 1.0$